

Exercise 3: Conditional probability and Central Limit Theorem

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Problem 1 (Expectation of the Geometric Distribution):

Let X be a random variable representing the number of trials needed to get the first success in a sequence of independent Bernoulli trials. The probability of success on each trial is p . This is a Geometric distribution with PMF

$$\mathbb{P}(X = k) = (1 - p)^{k-1}p \quad \text{for } k = 1, 2, 3, \dots$$

Compute the expected value of $\mathbb{E}[X]$.

Hint: Use the law of total expectation.

Problem 2 (The Memoryless Property of the Exponential Distribution):

The exponential distribution is often used to model the lifetime of a component (e.g., a lightbulb) or the waiting time for an event (e.g., the next customer arriving). A key feature is the **memoryless property**, which states that the future lifetime of the component does not depend on its current age.

Let $X \sim \text{Exp}(\lambda)$. Prove that for any $s, t \geq 0$ it holds

$$\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t).$$

Problem 3 (The Broken Stick):

A stick of length $L = 1$ is broken at a random point, chosen uniformly along its length. Let this first break point be X . We now have two pieces. Take the **longer** of these two pieces and break it again at a random point chosen uniformly along its length.

What is the expected length of the resulting piece that you are left with from this second break?

Problem 4 (The Insurance Company's Risk):

An insurance company has sold $n = 4000$ identical health insurance policies. From historical data, the average annual claim per policy is $\mu = 550$ CHF, with a standard deviation of $\sigma = 1200$ CHF. The high standard deviation suggests a skewed distribution (not normal). The company needs to determine how much money to hold in reserve.

- (a) Let S_{4000} be the total amount of claims from all 4000 policies. What are the expected value and the standard deviation of the total claims, $\mathbb{E}[S_{4000}]$ and $\text{SD}(S_{4000})$?
- (b) Use the Central Limit Theorem to estimate the probability that the total claims for the year will exceed 2.3 million CHF.
- (c) What is the probability that the average claim per policy is between 520 CHF and 580 CHF?

Problem 5 (De Moivre-Laplace Theorem (HARD)):

The De Moivre-Laplace theorem is a special case of the Central Limit Theorem, showing that the binomial distribution can be approximated by a normal distribution for a large number of trials.

Let $X_n \sim \text{Binomial}(n, p)$ and let $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ be the standard normal PDF. For a sequence of integers k_n such that the standardized value $x_{k_n} = \frac{k_n - np}{\sqrt{np(1-p)}}$ remains in a bounded set, prove the following limit statement:

$$\lim_{n \rightarrow \infty} \frac{\mathbb{P}(X_n = k_n)}{\frac{1}{\sqrt{np(1-p)}} f(x_{k_n})} = 1.$$

This means that the ratio of the true binomial probability to its normal approximation approaches 1.

*Hint: The key tools are **Stirling's approximation** $m! \approx \sqrt{2\pi m}(m/e)^m$ and the **Taylor expansion** $\ln(1+u) \approx u - u^2/2$. Structure your proof by analyzing the ratio directly.*